Fiber accelerometer — Solution

Y26 (2026) — English translation

A1 **0.20 pts**

Express the change in diameter of the winding region δD in terms of D, h and δh.

Let's express the volume of rubber:

$$V = \pi D^2 h$$

Let's differentiate the expression:

$$2D\,\delta D\,h + D^2\,\delta h = 0$$

From here:
Answer: \textbfAnswer:

$$\delta D = -\frac{D}{2h}\delta h$$

A2 **0.10 pts**

The change in total length of wire δL is proportional to δh

$$\delta L = \Lambda \delta h.$$

Express Λ in terms of N, D, h. The number of revolutions N is strictly fixed.

$$\delta L = \pi N \delta D = -\frac{\pi N D}{2h}\delta h$$

Answer: \textbfAnswer:

$$\Lambda = -\frac{\pi N D}{2h}$$

A3 **0.60 pts**

Find the strain energy of the wire W. Express your answer using N, D, h, d, E and δh

The volumetric strain energy density is expressed as:

$$w = \frac{E\delta L^2}{2L^2}$$

Wire volume:

$$V = \frac{\pi^2 d^2 N D}{4}$$

Let us express the answer in terms of the quantities specified in the condition:
Answer: \textbfAnswer:

$$W = \frac{E\pi^2 N D d^2 \delta h^2}{32h^2}$$

A4 — 0.50 pts

Let at some moment of time the body is displaced ($y \neq 0$) and the load is not in equilibrium ($x \neq y$). Obtain the equation of motion of the load as

$$\ddot{x} + \omega_0^2 x = Ay.$$

Express ω_0 and A in terms of M, h, N, D, d and E.

The total energy is equal to:

$$2W = \frac{E\pi^2 NDd^2(x-y)^2}{16h^2}$$

Let us express the force acting on the load:

$$F = \frac{\partial(2W)}{\partial(\delta h)} = M\ddot{x}$$

The final equation:

$$\ddot{x} + \frac{E\pi^2 NDd^2 x}{8Mh^2} = \frac{E\pi^2 NDd^2 y}{8Mh^2}$$

From here:

Answer: \textbfAnswer:

$$\omega_0 = \sqrt{\frac{E\pi^2 NDd^2}{8Mh^2}}$$

$$A = \frac{E\pi^2 NDd^2}{8Mh^2}$$

A5 — 0.50 pts

Express L_0 in terms of y_0, ω/ω_0, Λ and ζ.

Let's use the method of complex amplitudes:

$$x_0(-\omega^2 + 2\zeta i\omega\omega_0 + \omega_0^2) = \omega_0^2 y_0$$

We express $h_0 = x_0 - y_0$:

$$h_0 = y_0\left(-1 + \frac{\omega_0^2}{\omega_0^2 - \omega^2 + 2\zeta i\omega\omega_0}\right)$$

Taking into account that $L_0 = \Lambda h_0$, we write the answer:

Answer: \textbfAnswer:

$$L_0 = \Lambda y_0\left(-1 + \frac{1}{1 - \frac{\omega^2}{\omega_0^2} + 2\zeta i\frac{\omega}{\omega_0}}\right) = \Lambda y_0 \frac{-\frac{\omega^2}{\omega_0^2} + 2i\zeta\frac{\omega}{\omega_0}}{1 - \frac{\omega^2}{\omega_0^2} + 2i\zeta\frac{\omega}{\omega_0}}$$

A6 — 1.00 pts

Indicate the correspondence between frequency ranges and sensor operating modes. Express K_s and K_a in terms of M, h, N, D, d and E.

Let us consider the behavior of the load in the low-frequency limit. Let us consider one harmonic of oscillations y. Let us simplify the expression for L_0 in the limit $\zeta\omega_0 \ll \omega \ll \omega_0$:

$$L_0 \approx \frac{\Lambda\omega^2 y_0}{\omega_0^2},$$

which corresponds to $L = -\frac{\Lambda\ddot{y}}{\omega_0^2}$. When adding several low-frequency harmonics, this relationship still holds. Let us express K_a:

$$K_a = -\frac{\omega_0^2}{\Lambda} = \frac{E\pi d^2}{4Mh}$$

Now consider the behavior of the load in the high-frequency limit. Let us similarly consider one harmonic of oscillations y. Let us simplify the expression for L_0 in the limit $\omega \gg \omega_0$:

$$L_0 \approx -\Lambda y_0$$

When adding several high-frequency harmonics, this relation is still satisfied. In this case $K_s = -\dfrac{1}{\Lambda} = \dfrac{2h}{N\pi D}$.

Answer: \textbfAnswer: With $\zeta\omega_0 \ll \omega \ll \omega_0$ the sensor works as an accelerometer, and with $\omega \gg \omega_0$ – as a seismograph.

$$K_s = \frac{2h}{N\pi D}$$

$$K_a = \frac{E\pi d^2}{4Mh}$$

A7 — 0.40 pts

Calculate the values of K_a and ω_0.

Answer: \textbfAnswer:

$$K_a = 3.44 \cdot 10^4 \text{ s}^{-2}$$

$$\omega_0 = 2750 \text{ rad/}$$

B1 — 1.00 pts

Express the intensity of light I incident on the photodiode in terms of δL, I_0, λ, n, r and φ_0. Here, φ_0 is the incident phase difference between the light passing through the fiber wound on the right and left supports, respectively, at $\delta L = 0$. Note: You don't have to follow the φ_0 sign. In other words, solutions differing by the change $\varphi_0 \to -\varphi_0$ are considered equivalent.

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After passing through the interferometer, waves with complex amplitudes

$$\frac{ri}{\sqrt{2}} E_0 e^{2ikn\,\delta L + i\varphi_0} \quad \text{и} \quad \frac{r}{\sqrt{2}} E_0 e^{-2ikn\,\delta L},$$

arrive at the splitter, so the total is

$$E = \frac{ri}{2} E_0 e^{2ikn\,\delta L + i\varphi_0} - \frac{ri}{2} E_0 e^{-2ikn\,\delta L} = -\frac{riE_0}{2}\left(e^{2ikn\,\delta L + i\varphi_0 + i\pi} + e^{-2ikn\,\delta L}\right) =$$

$$= -\frac{riE_0}{2}\left(e^{2ikn\,\delta L + i\varphi_0/2 + i\pi/2} + e^{-2ikn\,\delta L - i\varphi_0/2 - i\pi/2}\right) e^{i\varphi_0/2 + i\pi/2} = -riE_0 \cos\left(\frac{4\pi}{\lambda} n\delta L + \frac{\varphi_0}{2} + \frac{\pi}{2}\right)$$

$$I = EE^* = r^2 I_0 \sin^2\left(\frac{4\pi}{\lambda} n\delta L + \frac{\varphi_0}{2}\right)$$

Answer: \textbfAnswer:

$$I = r^2 I_0 \sin^2\left(\frac{4\pi}{\lambda} n\delta L + \frac{\varphi_0}{2}\right)$$

B2 — 1.50 pts

Determine the numerical values of Y and f.

The oscillation period of the picture is $T = 63$ ms $- 13$ ms $= 50$ ms, hence:

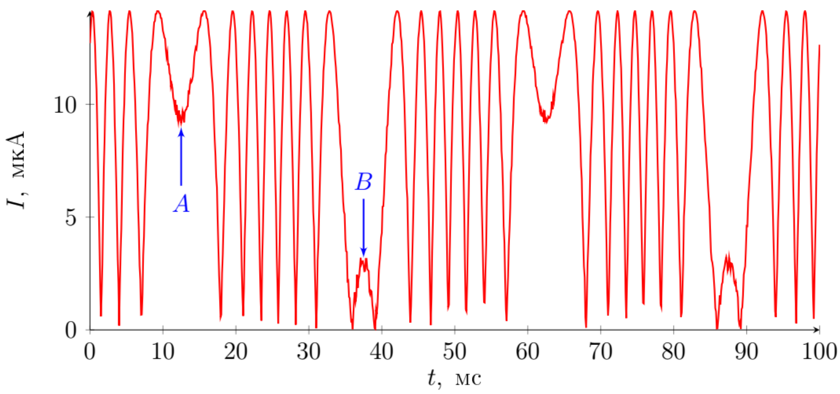
Half of the oscillation period corresponds to a change in $\ddot{y}$ from $-(2\pi f)^2 Y$ to $(2\pi f)^2 Y$.

Let us assume that the argument x of the function $\sin^2(x)$, which describes the light intensity at point A, is in the interval $(0, \pi/2)$. Then at the point with practically zero current to the left of B it is equal to 7π. The amplitude of current fluctuations is $I_o = 14.2$ μA. Current values at points A and B: $I_A = 9.7$ μA, $I_B = 3.2$ μA. That means $x_B - x_A = 7\pi + \arcsin(\sqrt{I_B/I_o}) - \arcsin(\sqrt{I_A/I_o}) = 21.5$. As a result,

$$x_B - x_A = \frac{4\pi}{\lambda}(\delta L_B - \delta L_A) = \frac{4\pi n}{\lambda} \cdot 2\frac{(2\pi f)^2 Y}{K_a} \quad \Rightarrow \quad Y = (x_B - x_A)\frac{\lambda K_a}{32\pi^3 n f^2} = 1.46\ \mu\text{m}$$

Answer: \textbfAnswer:

$$f = 20\ \text{Hz}$$



B3 — 0.30 pts

Using the theorem about the uniform distribution of energy over degrees of freedom, write down an expression for $\langle \delta L^2 \rangle$—fluctuations δL at room temperature. Express your answer in terms of Boltzmann's constant k_B, T, E, D, N and d.

The described system has one degree of freedom, but the deformation energy at the two supports adds up, therefore

$$2\langle W \rangle = \frac{1}{2}k_B T \quad \Rightarrow \quad \langle \delta h^2 \rangle = k_B T \frac{8h^2}{E\pi^2 N D d^2}$$

Taking into account that $\langle \delta L^2 \rangle = \Lambda^2 \langle \delta h^2 \rangle$, we obtain

$$\langle \delta L^2 \rangle = 2k_B T \frac{ND}{Ed^2}$$

Answer: \textbfAnswer:

$$\langle \delta L^2 \rangle = 2k_B T \frac{ND}{Ed^2}$$

B4 — 0.60 pts

If the temperature of this section of the optical fiber differs from room temperature by a small amount ΔT_i, then when light passes through it, phase $\varphi_i \neq \varphi_{i,0}$ occurs. Express $\Delta\varphi_i = \varphi_i - \varphi_{i,0}$ in terms of ΔT_i, ΔL_i, α, β, n and λ.

$$\Delta\varphi_i = \frac{2\pi}{\lambda}(n+\beta\Delta T_i)\Delta L_i(1+\alpha\Delta T_i) - \frac{2\pi}{\lambda}n\Delta L = \frac{2\pi}{\lambda}(n\alpha+\beta)\Delta L_i\Delta T_i$$

Answer: \textbfAnswer:

$$\Delta\varphi_i = \frac{2\pi}{\lambda}(n\alpha+\beta)\Delta L_i\Delta T_i$$

B5 — 0.70 pts

Express $\langle\Delta\Phi^2\rangle$ in terms of $\langle\Delta T^2\rangle$, ΔL, L, n, α, β and λ.

Let's calculate $\langle\Delta\Phi^2\rangle$ directly:

$$\langle\Delta\Phi^2\rangle = \left\langle\left(\sum\Delta\varphi_i\right)^2\right\rangle = \left(\frac{2\pi}{\lambda}(n\alpha+\beta)\Delta L\right)^2\left\langle\left(\sum\Delta T_i\right)^2\right\rangle$$

Using the properties ΔT_i,

$$\left\langle\left(\sum\Delta T_i\right)^2\right\rangle = \left\langle\sum\Delta T_i^2\right\rangle + \left\langle\sum\Delta T_i\Delta T_j\right\rangle = p\langle\Delta T^2\rangle = \frac{L}{\Delta L}\langle\Delta T^2\rangle.$$

Answer: \textbfAnswer:

$$\langle\Delta\Phi^2\rangle = \left(\frac{2\pi}{\lambda}(n\alpha+\beta)\right)^2 L\Delta L\langle\Delta T^2\rangle$$

B6 — 1.00 pts

Express $\langle\Delta\Psi^2\rangle$ in terms of $\langle\Delta\Phi^2\rangle$.

The temperature fluctuations described earlier are a much slower process than the propagation of light, so for any $\Delta\Phi$ it holds that

$$\Delta\Psi = 2\Delta\Phi \quad\Rightarrow\quad \langle\Delta\Psi^2\rangle = 4\langle\Delta\Phi^2\rangle$$

Note that the answer differs by a factor of 2 from propagation along a fiber of length $2L$!

Answer: \textbfAnswer:

$$\langle\Delta\Psi^2\rangle = 4\langle\Delta\Phi^2\rangle$$

B7 — 0.60 pts

Find the root-mean-square difference in the changes in the phase shift on the two arms of the interferometer $\langle(\Delta\Psi_1-\Delta\Psi_2)^2\rangle$. Express your answer using $\langle\Delta\Psi^2\rangle$.

The phase shift on both interferometers is independent, therefore

$$\langle(\Delta\Psi_1-\Delta\Psi_2)^2\rangle = \langle\Delta\Psi_1^2\rangle + \langle\Delta\Psi_2^2\rangle = 2\langle\Delta\Psi^2\rangle$$

Answer: \textbfAnswer:

$$\langle(\Delta\Psi_1-\Delta\Psi_2)^2\rangle = 2\langle\Delta\Psi^2\rangle$$

B8 — 1.00 pts

Consider that the sensor operates at $T = 305\ \mathrm{K}$ and the magnitude of temperature fluctuations $\langle\Delta T^2\rangle$ is $1.0\cdot10^{-6}\ \mathrm{K}^2$. What is the minimum acceleration $a_{\min}$ that can be measured using the proposed sensor?

Due to thermal fluctuations in the mechanical system, the length of the interferometer arms has a noise of

$$\sqrt{\langle \delta L^2 \rangle} = \sqrt{2k_B T \frac{ND}{Ed^2}} = 7.1 \cdot 10^{-12} \text{ m}$$

. Due to thermal fluctuations in the fiber, the phase difference has a noise of

$$\sqrt{\langle (\Delta\Psi_1 - \Delta\Psi_2)^2 \rangle} = \frac{4\pi\sqrt{2}}{\lambda}(n\alpha + \beta)\sqrt{\pi D N \Delta L} \cdot \sqrt{\langle \Delta T^2 \rangle} = 2 \cdot 10^{-3},$$

, which leads to noise in the calculated δL equal to $\frac{\lambda}{8\pi n} \cdot 2 \cdot 10^{-3} = 0.6 \cdot 10^{-10}$ m. Thus, the contribution from the second effect is an order of magnitude greater, and the minimum possible acceleration is:

Answer: \textbfAnswer:

$$a_{\text{min}} \approx K_a \cdot 0.6 \cdot 10^{-10} \text{ m} = 2 \cdot 10^{-6} \ m/s^2$$